

Reactive Photon Field Model (RPFM)

Author :

Eng. Adnan Abdulaziz
Independent Researcher.

Model Description

This model provides a circuit-theory-inspired representation of photon physics using simple electromagnetic parameters.

It reformulates photon behavior in terms of effective circuit analogies such as capacitance, inductance, and vector potential, enabling a more intuitive and computationally accessible framework for analysis.

The approach opens a new perspective for studying complex physical concepts—such as photon energy and effective mass—through simplified parameter mappings rather than full field-theoretic formalisms.

This framework also supports practical interpretation through laboratory-scale signal mapping, where voltage is defined as:

$$V_{\text{map}} = A_0 c$$

using vector potential amplitude A_0 and the speed of light c as the scaling bridge between field representation and measurable electrical signals.

This voltage mapping provides an opportunity for model validation, allowing theoretical predictions to be compared directly with measurable laboratory or simulation-based voltage signals.

Overview :

This project presents a computational and theoretical framework for modeling photon-like electromagnetic modes in free space using an LC-field analogy. The model connects electric and magnetic field storage through effective capacitance and inductance derived from vacuum properties and geometric confinement. The formulation is based on vector potential amplitude A_0 , vacuum impedance Z_0 , and spatial scaling through wavelength and effective radiating area. Both spherical and disk-like geometries are considered to represent different field configurations. The goal is to describe photon behavior as a structured electromagnetic mode with a unified field-based effective mass representation.

Equations:**Energy frequency mode :**

$$U = hc/\lambda = hc / 2\pi R, \quad w = c/R \quad \text{thus} \quad hf = wh / 2\pi = hf$$

$$w = c / R, \quad f = c / 2\pi R = c / \lambda$$

• Geometry core :

$$R = \frac{\lambda}{2\pi}$$

$$\lambda = 2\pi R$$

$$k = \frac{1}{R}$$

• Vector potential :

$$A_0 = BR$$

$$A_0 = \frac{B\lambda}{2\pi}$$

• Voltage mapping :

$$V = ER$$

$$V = cA_0$$

$$V = E\lambda$$

• Effective areas :

$$A_{\text{cross}} = \pi R^2$$

$$A_{\text{sphere}} = 4\pi R^2$$

• Curvature :

$$K = \frac{1}{A_{\text{eff}}}, \quad K = \frac{1}{\pi R^2} = \frac{k^2}{\pi}, \quad K = \frac{1}{4\pi R^2} = \frac{k^2}{4\pi}$$

• Energy :

$$U = \frac{1}{2}\epsilon_0 E^2 AL + \frac{1}{2} \frac{B^2}{\mu_0} AL$$

$$U = \epsilon_0 E^2 AL$$

$$U = \frac{B^2}{\mu_0} AL$$

• **Mass :**

Inductive $m_{eff} = (\mathcal{L}_{eff} - 4 \pi^2 A_0^2) / Z_0^2$

Capacitive $m_{eff} = C_{eff} - 4 \pi^2 A_0^2$

LC analogy :

$$C_{eff} = \epsilon_0 \frac{A_{eff}}{L_{eff}}$$

$$L_{eff} = \mu_0 \frac{L_{eff}}{A_{eff}}$$

• **Free space impedance :**

$$Z_0 = \sqrt{\frac{L_{eff}}{C_{eff}}} = \sqrt{\frac{\mu_0}{\epsilon_0}}$$

• **Quantum Scaling Relation :**

$$\frac{h}{\lambda^2} \sim \text{spectral scaling term}$$

$$\text{for } A_{eff} = A_{eff} = 4 \pi R^2$$

$$\frac{h}{\lambda^2} = \frac{4\pi A_0^2}{Z_0} \text{ kg/s}$$

$$h/\lambda^2 = (\pi / Z_0) A_0^2 \text{ (kg/s) } \quad \text{for } A_{eff} = \pi R^2$$

Core Objective :

The goal of this project is to investigate whether electromagnetic wave behavior can be consistently represented through a structured field-based framework.

The study focuses on:

- Geometry-based scaling of wave structure
- Field amplitude relationships governed by vector potential
- Wave-number dependent transformations
- Energy and flux distribution patterns in free space
- Effective analogies inspired by LC-like electromagnetic systems

This framework is intended for numerical exploration, visualization, and hypothesis testing of photon-like electromagnetic modes in both spherical and disk geometries.

Project Structure :

code/ → Core simulation modules and numerical models

plots/ → Generated visualizations and simulation outputs

requirements.txt → Python dependencies for reproducibility

README.md → Project overview and usage description

THEORY.md → Mathematical and conceptual framework of the model.

Simulation Modules :

The toolkit includes the following computational components:

- Wave behavior simulation module
- Geometric scaling and curvature analysis module
- Electromagnetic energy distribution module
- Flux and flow dynamics module
- LC-inspired analogy modeling module

Each module is designed to isolate a specific aspect of the system, enabling independent analysis, visualization, and validation of the underlying assumptions.

Outputs :

The project generates sample output images from the implemented simulation codes. These images represent the visual behavior of the model under different parameters and initial conditions.

All output images are stored in the `plots/` directory for reference and demonstration purposes.

Design Philosophy :

This project follows a research-oriented computational design approach focused on clarity, modularity, and interpretability.

Key principles include:

- Modular decomposition of physical and mathematical concepts.
- Independent validation of each model component.
- Emphasis on reproducibility and transparent computation.
- Prioritization of visual and geometric interpretability of results.

The framework is intentionally structured to support incremental development, parameter exploration, and future theoretical extensions.

Installation :

1 -Install the required dependencies using:

```
pip install numpy scipy matplotlib pyqt5
```

2- Create virtual environment and activate it.

```
python -m venv venv
```

```
source venv/bin/activate # Mac/Linux
```

```
venv\Scripts\activate # Windows
```

Usage :

Each simulation script in this project is standalone and can be executed independently.

Run any module using:

```
python filename.py
```

Intended Use :

This project is intended for:

- Computational physics exploration
- Electromagnetic wave modeling studies
- Research prototyping and hypothesis testing
- Educational visualization of wave-based systems

It is not intended as a replacement for established physical theory, but rather as a complementary computational framework for exploratory analysis and conceptual development.

Future Development :

Planned extensions of this project include:

- Interactive simulation interface for real-time exploration
- Real-time parameter control system for dynamic tuning of models
- Expanded multi-scale modeling across different geometric regimes
- Integration with experimental or reference datasets for validation
- Enhanced visualization pipeline for higher-dimensional field analysis

Collaboration :

This project is open for academic and research collaboration in the following areas:

- Computational electromagnetics
- Wave physics modeling and analysis
- Applied mathematical physics
- Numerical simulation and field-based modeling systems

Researchers interested in extending, testing, or validating the framework are welcome to contribute and engage in further development of the model.

License :

This project is released for academic and research use.

It is intended for non-commercial exploration, study, and scientific development. Further usage, modification, or extension for research purposes is encouraged.

Author Note :

This framework is developed as a structured computational exploration of wave-field behavior using geometry-driven and field-coupled modeling principles.

It is intended as an evolving research tool for studying electromagnetic wave structure through numerical simulation and conceptual modeling.

